OAI VWL 20.11.25

Donnerstag, 20. November 2025 08:26

Blatt 2:

2. $y = 50 \cdot K^{1/2}$

a.) $\Pi = 50 \cdot K^{1/2} - (1+r) \cdot K$

$\frac{d\Pi}{dK} \stackrel{!}{=} 0 : 25 \cdot K^{-1/2} - (1+r) = 0 \Rightarrow 25 \cdot K^{-1/2} = 1+r$

$\hat{=} MPK \quad \hat{=} MC$

$K^{-1/2} = \frac{1+r}{25}$

$K' = \left(\frac{25}{1+r}\right)^2$

b.) $r = 0{,}02$

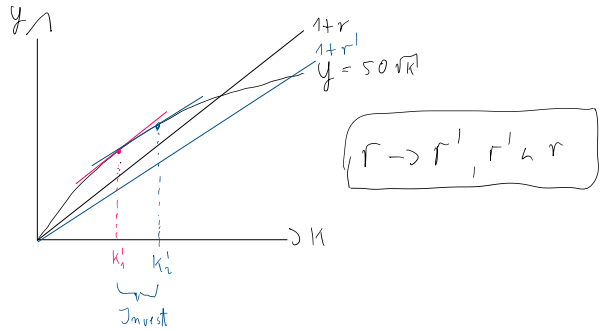
$K_1' = \left(\frac{25}{1{,}02}\right)^2 = 600{,}7$

c.) $r = 0{,}04$

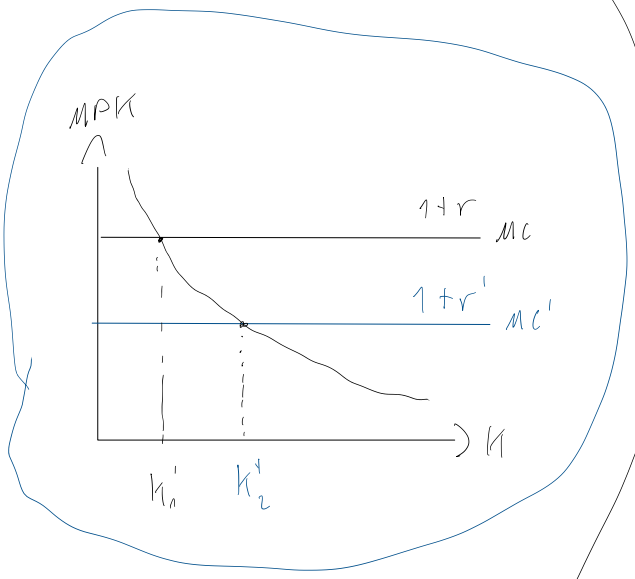
$K_2' = \left(\frac{25}{1{,}04}\right)^2 = 577{,}8$

$i = K_2' - K_1' = -22{,}9$

d.)



$r \to r', r' < r$



3. $y = 2K^{2/3}$ $r = 0{,}03$

a.) $\Pi = 2 \cdot K^{2/3} - (1+r) K$

$\frac{d\Pi}{dK} \stackrel{!}{=} 0 : \frac{2}{3} \cdot 2 \cdot K^{-1/3} - (1+r) = 0$

$\Rightarrow \frac{2}{3} \cdot 2 \cdot K^{-1/3} \cdot \frac{K}{K} = 1+r$

$\Rightarrow \frac{2}{3} \cdot 2 \cdot K^{2/3} \cdot \frac{1}{K} = 1+r$

$\Rightarrow \frac{2}{3} \cdot y \cdot \frac{1}{K} = 1+r$

$\Rightarrow K = \frac{2/3}{1+r} \cdot y$ $\quad \frac{2/3}{1+r} = \alpha$

$K = \alpha \cdot y$

b.) $y_1 = 1000$

$y_2 = 1500$

$i = K_2' - K_1'$

$= \alpha \cdot y_2 - \alpha \cdot y_1$

$= \alpha \cdot (y_2 - y_1)$

$\Rightarrow I = \frac{2/3}{1{,}03} \cdot 500$

$= 323{,}6$

c.) $400 = \alpha \cdot (y_3 - y_2)$

$400 \cdot \frac{1{,}03}{2/3} = y_3 - y_2$

$y_3 - y_2 = 618$

$y_2 = 1500 \Rightarrow y_3 = 2118$

Blatt 3

1. $s_n = a_0 \cdot \frac{1 - q^{n+1}}{1 - q}$ $\lim_{n \to \infty} q^{n+1} = 0$, da $1 > q > 0$

a.) $a_0 = 15\,000$

$n = 3$

$q = (1 - r) = 0{,}95$

$s_3 = 15\,000 \cdot \frac{1 - 0{,}95^{3+1}}{1 - 0{,}95}$

$= 55\,648{,}13$

b.) $n = \infty$

$s_\infty = a_0 \cdot \frac{1}{1 - q}$

hier: $15\,000 \cdot \frac{1}{1 - 0{,}95} = 300\,000$

c.) $r = 0{,}11$; $q = 0{,}89$

$n = 5$

$s_5 = 70\,283{,}85$

d.) $r = 0{,}1$

$n = \infty$

$s_\infty = 150\,000$